

AHB to APB Bridge Design Specification

AHB (Advanced High-Performance Bus): -

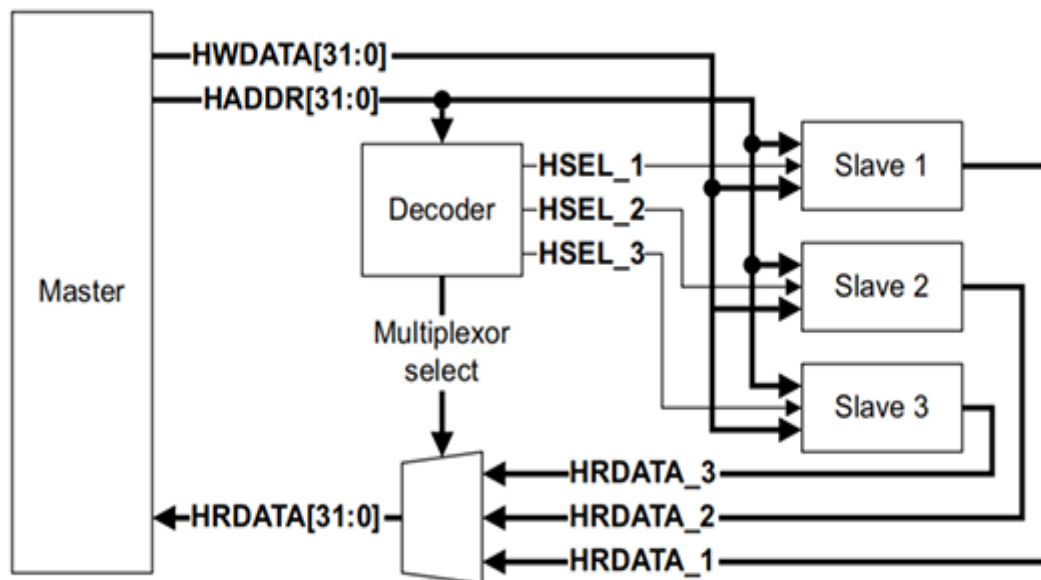


Fig-1: AHB Block Diagram

AMBA AHB is a bus interface designed for high-performance, synthesizable designs. It defines the connection between various components, including masters, interconnects, and slaves. AHB incorporates features essential for high-speed, high-frequency systems, such as:

- Burst data transfers
- Operation on a single clock edge
- Non-tristate implementation
- Support for wide data bus configurations, including 64, 128, 256, 512, and 1024 bits

AHB slaves typically include internal memory devices, external memory interfaces, and high-bandwidth peripherals. While low-bandwidth peripherals can also function as AHB slaves, they are usually placed on the AMBA Advanced Peripheral Bus (APB) to optimize system performance. An APB bridge, acting as an AHB slave, facilitates communication between AHB and APB.

A typical single-master AHB system consists of an AHB master and multiple AHB slaves, interconnected by a bus system that includes an address decoder and a slave-to-master multiplexor. The address decoder determines which slave is selected based on the master's address, while the multiplexor directs the selected slave's data output back to the master.

For systems requiring multiple masters, AHB supports multi-master configurations using an interconnect component that manages arbitration and signal routing, ensuring communication between masters and the appropriate slaves.

AHB Master Interface: -

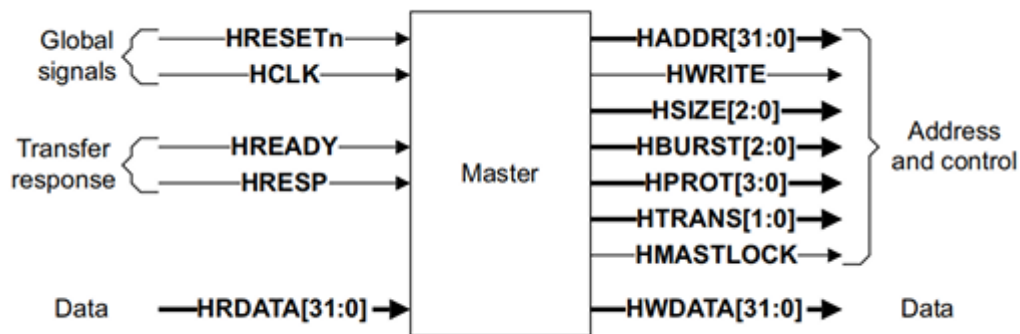


Fig-2: AHB Master Interface

A master initiates read and write operations by providing address and control information. Figure 2 illustrates the master interface.

AHB Slave Interface: -

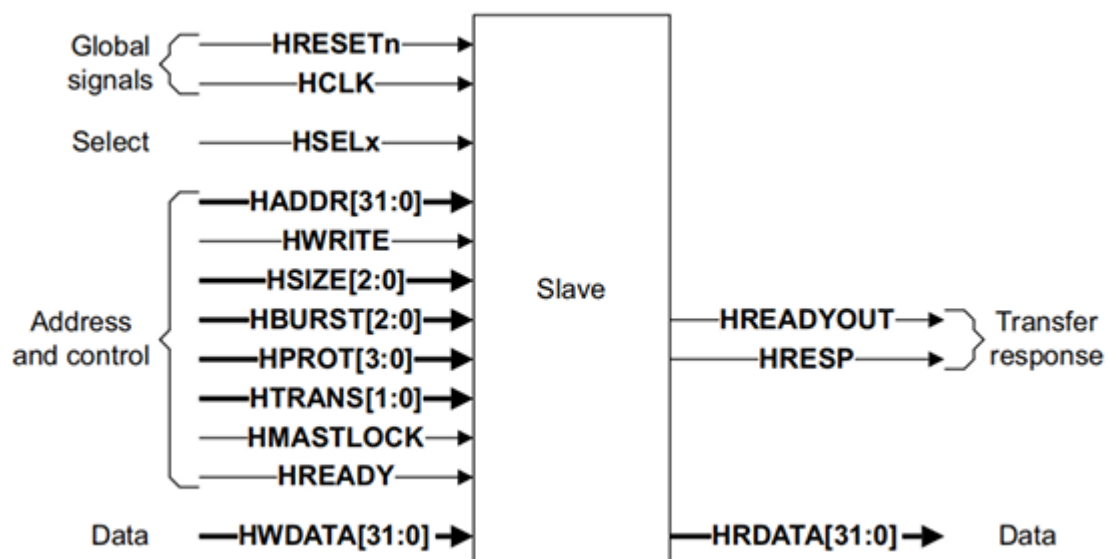


Fig-3: AHB Slave Interface

A slave, on the other hand, responds to transfer requests from masters within the system. It utilizes the HSELx select signal from the decoder to determine when to engage in a bus transfer. The slave then communicates back to the master regarding:

- The completion or extension of the bus transfer
- Whether the transfer was successful or failed

Figure 3 depicts the slave interface.

APB (Advanced Peripheral Bus): -

The Advanced Peripheral Bus (APB) is a component of the Advanced Microcontroller Bus Architecture (AMBA) protocol family. It provides a cost-effective interface designed for low power consumption and simplified connectivity. Unlike pipelined protocols, APB is optimized for low-bandwidth peripherals that do not require the high performance of the AXI protocol.

APB aligns signal transitions with the rising edge of the clock, making it easier to integrate APB peripherals into various design flows. Each data transfer requires a minimum of two clock cycles.

APB can interface with:

- AMBA Advanced High-performance Bus (AHB)
- AMBA Advanced High-performance Bus Lite (AHB-Lite)
- AMBA Advanced Extensible Interface (AXI)
- AMBA Advanced Extensible Interface Lite (AXI4-Lite)

It is commonly used to access programmable control registers of peripheral devices.

AHB to APB Bus: -

Importance: -

The AHB to APB bridge functions as an AHB slave, facilitating communication between the high-speed AHB and the low-power APB. It translates read and write operations from the AHB into corresponding transactions on the APB. Since the APB is non-pipelined, wait states are introduced during data transfers when the AHB must pause for the APB to complete operations. This bridge is essential for connecting low-bandwidth APB peripherals with high-bandwidth ARM processors and other high-speed AHB devices, ensuring seamless data transmission without loss. The AHB2APB interface captures addresses, control signals, and data from the AHB, manages APB peripherals, and transmits response signals and data back to the AHB. Designed to function regardless of differences in frequency and phase between AHB and APB clocks, the AHB2APB bridge enables data transfer from AHB to APB during write cycles and from APB to AHB during read cycles. It serves as a critical link between the AMBA high-performance bus (AHB) and the AMBA peripheral bus (APB), ensuring proper latching of address, control, and data signals for APB peripherals.

Architecture: -

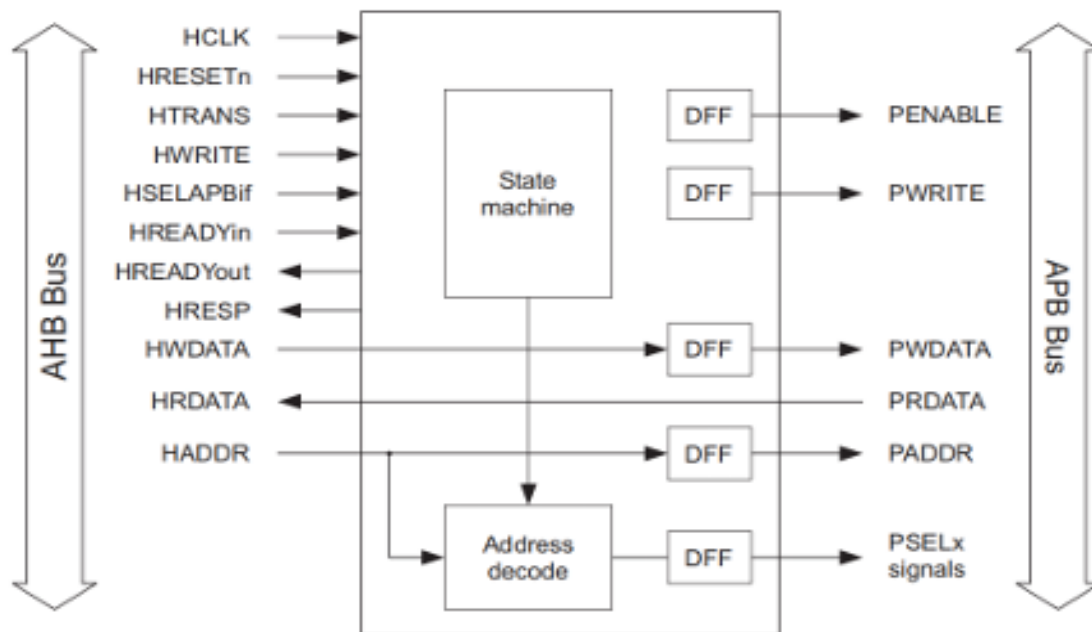


Fig-4: AHB to APB Bridge Architecture

This module consists of three main sections:

- **AHB slave bus interface**
- **APB transfer state machine**, which operates independently of the device memory map
- **APB output signal generation**

To integrate new APB peripherals or modify the system memory map, only the address decoding sections need to be updated. The base addresses of peripherals—such as the timer, interrupt controller, and remap and pause controller—are defined within the AHB to APB bridge interface. The bridge selects the appropriate peripheral based on its base address, and the entire APB address range is configured within the bridge. These base addresses can be implementation-specific. The APB peripheral standard defines only the register offsets (relative to an unspecified base address), the meaning of register bits, and the minimum required functionality.

The APB data bus is divided into two unidirectional paths:

- **Read path (PRDATA)** – Data flows from the peripherals to the bridge.
- **Write path (PWDATA)** – Data moves from the bridge to the peripherals.

This separation simplifies bus management by eliminating turnaround delays between the bridge and peripherals. In a typical system, where the bridge is the sole bus master, PWDATA is driven continuously. PRDATA, on the other hand, is a multiplexed signal combining outputs

from all peripherals on the bus and is only actively driven when the bridge selects a slave for an APB read operation.

While it is technically feasible to merge these two data buses into a single bidirectional bus, careful design considerations are necessary to prevent potential conflicts between the bridge and the peripherals.

Signal Descriptions: -

AHB Interface Signals (Slave Side)

Signal	Direction	Description
HCLK	Input	AHB system clock. It synchronizes all AHB transfers.
HRESETn	Input	Active-low reset signal for the AHB bus. Resets the bridge logic.
HADDR	Input	32-bit address bus from the AHB master, specifying the target peripheral.
HTRANS	Input	2-bit transfer type signal (IDLE, BUSY, NONSEQ, SEQ).
HWRITE	Input	Write enable signal. High (1) for write, low (0) for read.
HSIZE	Input	3-bit transfer size (Byte, Halfword, Word).
HBURST	Input	3-bit burst type (SINGLE, INCR, WRAP).
HPROT	Input	4-bit protection control signal for privileged/user and cacheable/bufferable attributes.
HWDATA	Input	32-bit write data bus carrying data from AHB master to the bridge.
HRDATA	Output	32-bit read data bus carrying data from the bridge to the AHB master.
HREADY	Output	Indicates whether the bridge is ready to accept a new transfer.
HRESP	Output	2-bit response signal (OKAY, ERROR, etc.).

APB Interface Signals (Master Side)

Signal	Direction	Description
PCLK	Input	APB system clock.
PRESETn	Input	Active-low reset signal for APB. Resets the APB peripherals.

Signal	Direction	Description
PADDR	Output	32-bit address bus for selecting the APB peripheral.
PWRITE	Output	Write enable signal. High (1) for write, low (0) for read.
PENABLE	Output	Enables the APB transfer.
PWDATA	Output	32-bit write data bus carrying data from the bridge to APB peripherals.
PRDATA	Input	32-bit read data bus carrying data from APB peripherals to the bridge.
PSELx	Output	Peripheral select signal (PSEL[0], PSEL[1], etc.), indicating which APB peripheral is active.
PREADY	Input	Indicates whether the selected APB peripheral is ready for the transfer.
PSLVERR	Input	Peripheral error response signal. High (1) indicates an error.

AHB to APB Bridge Components: -

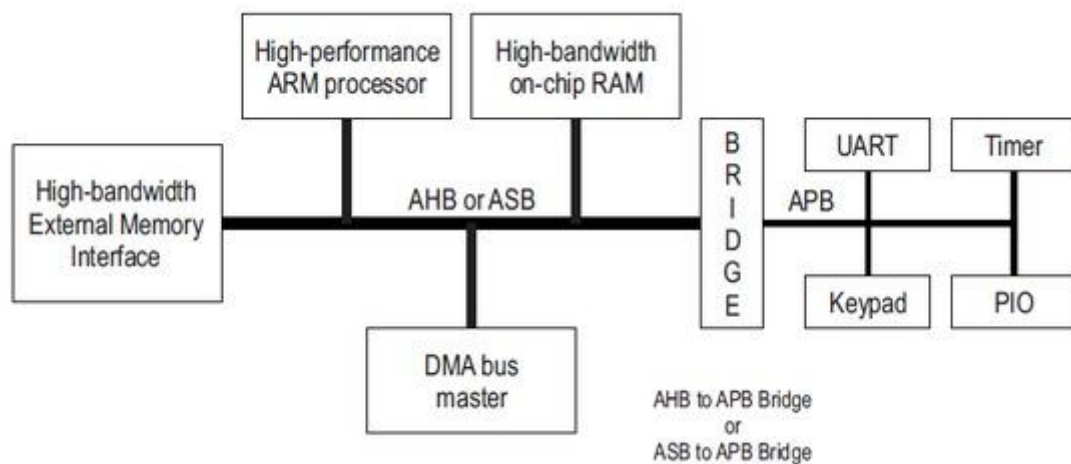


Fig-5: AHB to APB bridge Block Diagram

The AHB Slave Bus Interface utilizes the standard AHB slave bus interface, which includes:

- Valid transfer detection logic, responsible for identifying when a valid transfer is accessing the slave.
- Address and control registers, which store information from the address phase of the transfer for use during the data phase.

Since the timing of read and write transfers differs between AHB and APB, either the current or the previously latched address value is required to ensure the correct generation of APB transfers. To handle this, a multiplexer is used to select between the current address input for read operations and the registered address for write operations.

The APB transfer state machine controls how data is transferred between the AHB to APB bridge and the APB peripherals. Since APB is a non-pipelined bus, transactions follow a simple two-phase protocol: setup and access. The state machine ensures that read and write transfers are executed correctly while adhering to APB timing requirements.

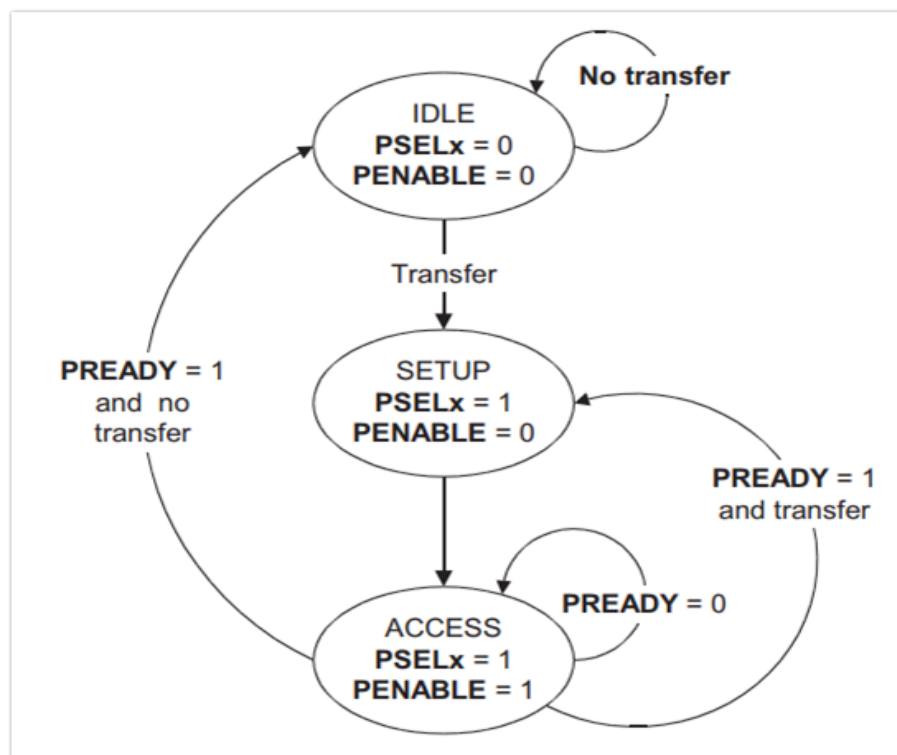


Fig-6: APB Transfer State Machine

The APB transfer state machine typically consists of the following states:

1. IDLE State

- The default state where no transfer is in progress.
- PSELx = 0, PENABLE = 0.
- The state machine remains here until an AHB transfer is detected.

2. SETUP State

- The start of an APB transfer.

- The bridge drives the **peripheral select (PSELx)** signal high to indicate which APB peripheral is being accessed.
- The address (PADDR), write data (PWDATA), and control signals (PWRITE) are set.
- PENABLE remains low in this state.

3. **ACCESS State**

- The actual data transfer takes place in this state.
- PENABLE is asserted (1), indicating that the peripheral should capture the data or provide a response.
- The peripheral acknowledges the transfer via PREADY.
- If PREADY is high:
 - Read (PRDATA is captured) or write (PWDATA is sent) completes.
 - The state machine transitions back to **IDLE** (if a single transfer) or remains in **SETUP** for the next transfer.
- If PREADY is low:
 - The state machine remains in **ACCESS** until PREADY goes high.

The AHB output signals consist of the following:

- HRDATA is directly driven by the current value of PRDATA. Since APB slaves only assert PRDATA during the enable phase of an APB transfer and keep it LOW at other times, bus contention is avoided on HRDATA, assuming an OR-based bus connection for both AHB and APB read data buses.
- HREADYout is generated using a registered signal to enhance output timing. The APB bridge inserts wait states during the ST_READ and ST_WRITEP states, as well as in the ST_WENABLEP state when the next transfer to be executed is a read operation.
- HRESP remains LOW at all times, as the APB bridge does not generate SPLIT, RETRY, or ERROR responses.